

Daily Review

Booklet p.24-25 — Light Years

- 1 Define a light year.
- 2 State the speed of light in a vacuum.
- 3 A star is 5.0 light years away. Distance in metres?
- 4 Why are distances in space measured in light years?
- 5 How many seconds in one year?

The distance light travels in a year

$$3.0 \times 10^8 \text{ m s}^{-1}$$

$$4.7 \times 10^{16} \text{ m}$$

The numbers in metres are huge

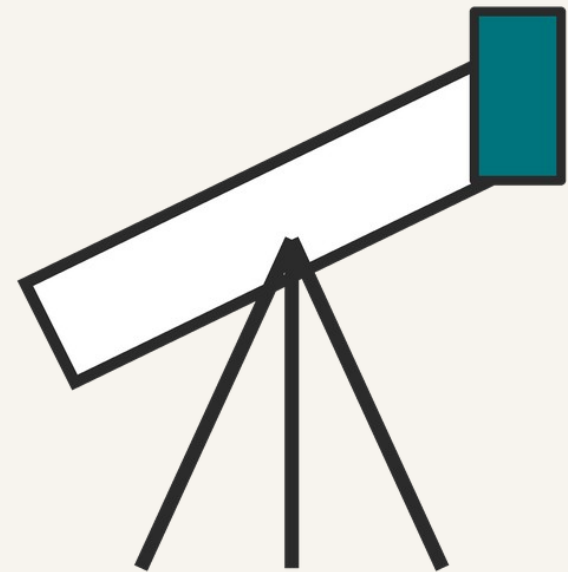
$$3.15 \times 10^7 \text{ s}$$

SPACE

Telescopes

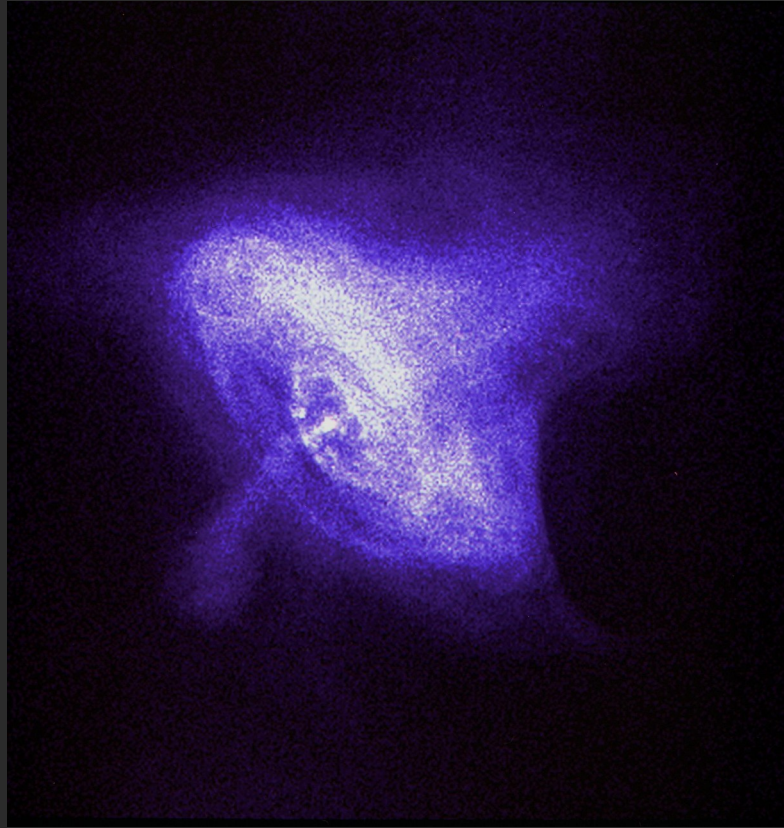
Every part of the spectrum tells us something different.

National 5 Physics (S3)



THE BIG QUESTION

This cloud looks completely different in X-rays, infrared and visible light. Why look at it in all of them?



The Crab Nebula in X-rays — NASA

PART 1

Telescopes

Why we look at the sky in far more than visible light.

By the end of this part...

1

I can describe how different parts of the electromagnetic spectrum give us information about objects in space

Show me — Describe each word

SHOW ME BOARDS

1 telescope

An instrument for seeing
distant objects

2 electromagnetic spectrum

The whole family of
radiations

3 wavelength

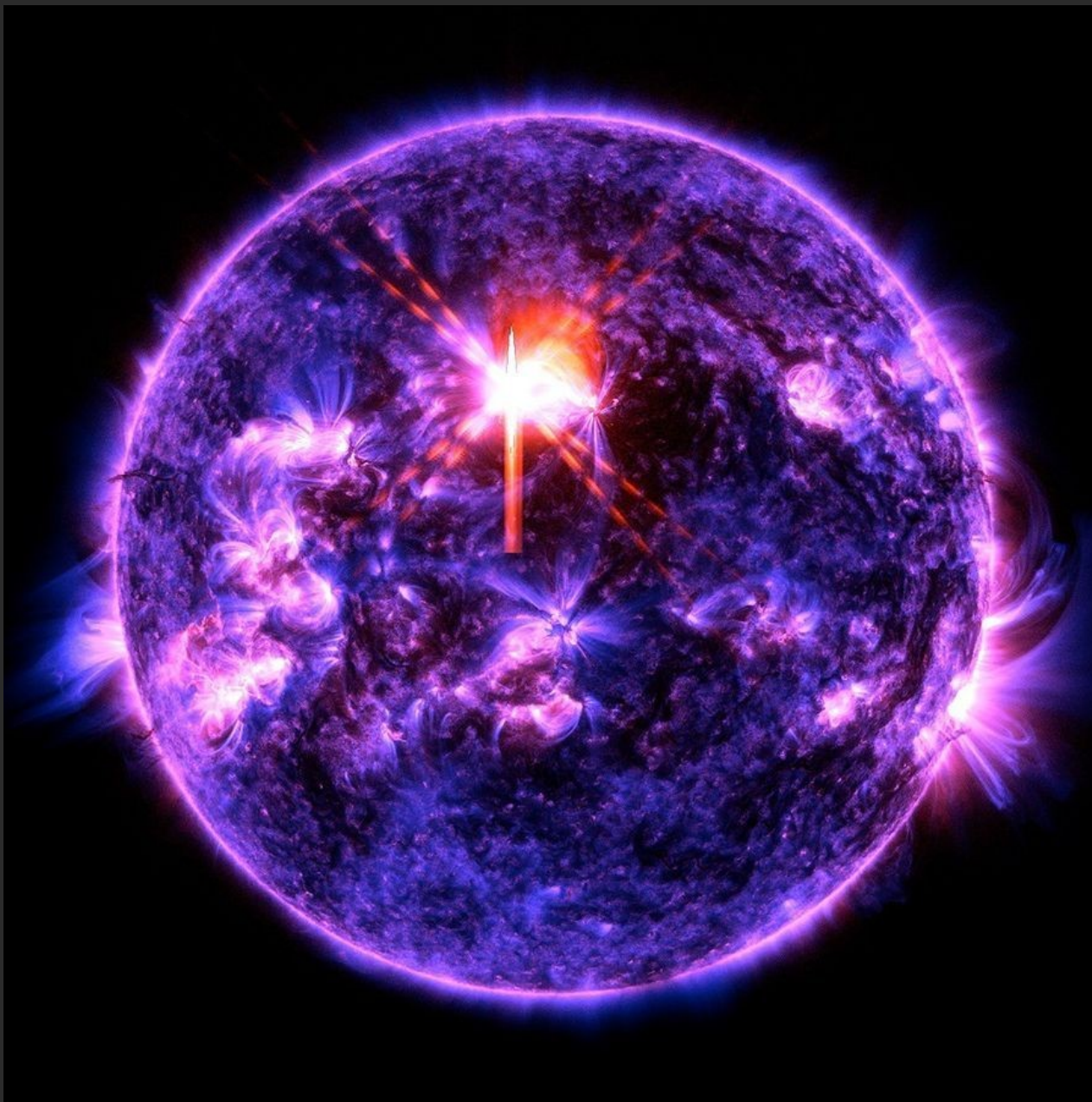
The length of one wave

4 infrared

Radiation just beyond
red light

5 radio waves

The longest wavelengths
of all



The Sun in ultraviolet — NASA / Solar Dynamics Observatory (SDO)

The electromagnetic spectrum

Booklet p.26



radio

microwave

infrared

visible

ultraviolet

X-ray

gamma

longest wavelength

shortest wavelength

- All of them travel at $3.0 \times 10^8 \text{ m s}^{-1}$ in a vacuum.

Task 1 – order the spectrum

Booklet p.26

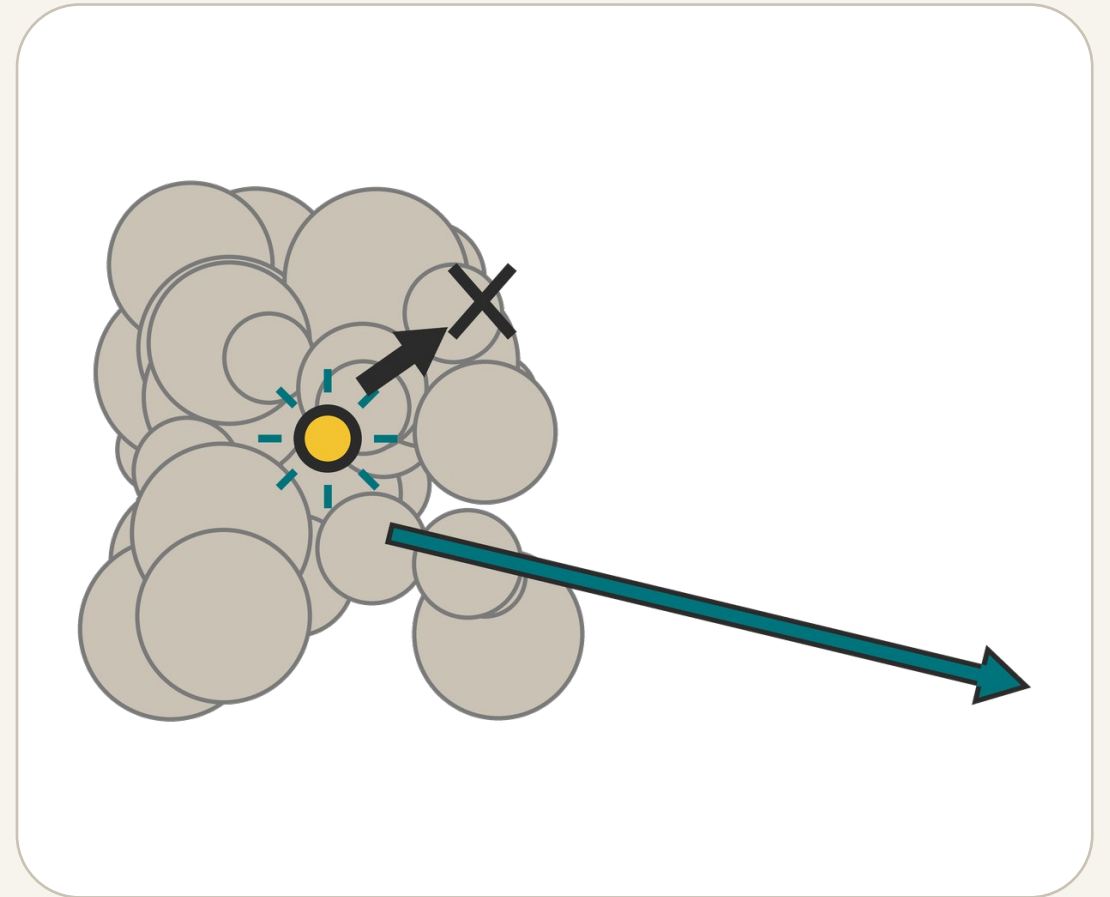
Fill in the spectrum, from the longest wavelength to the shortest.

Booklet p.26, Task 1

Why infrared sees through dust

Booklet p.26, Task 2

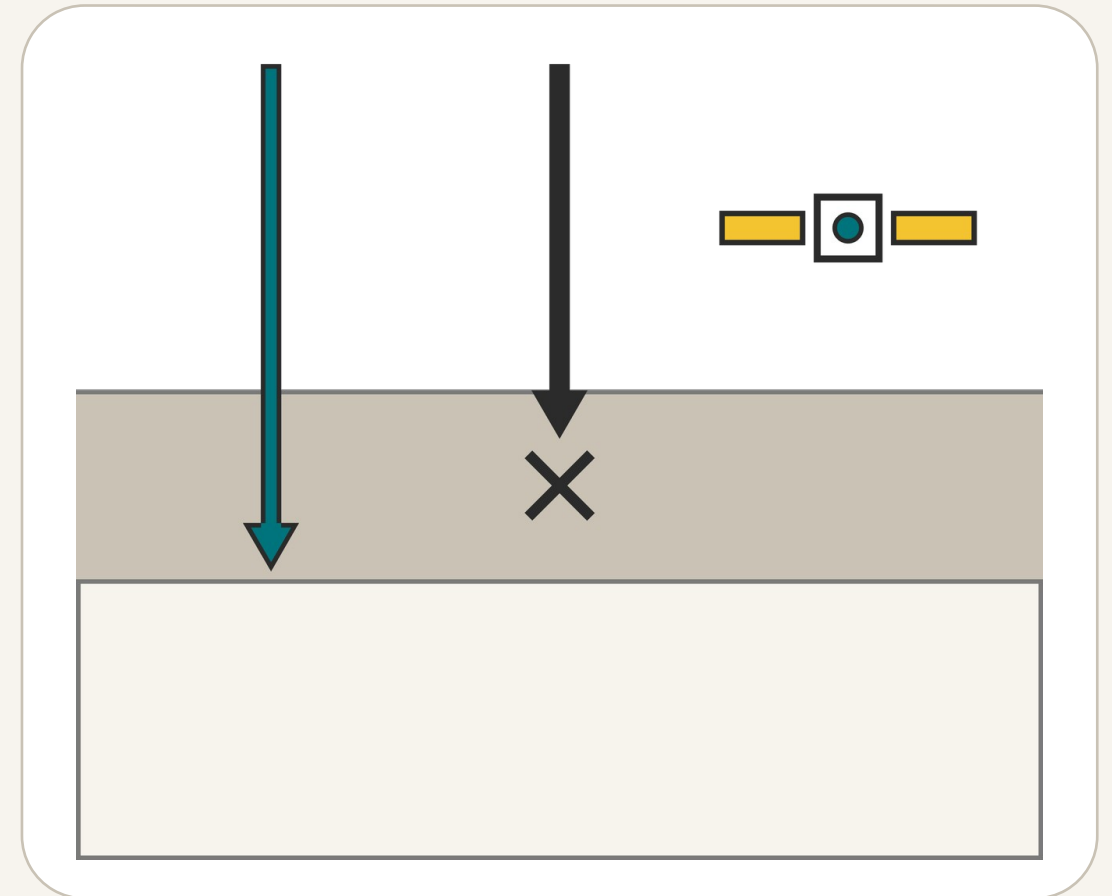
- Objects in space give out many kinds of radiation.
- Dust grains scatter and absorb visible light.
- Infrared has a longer wavelength and gets through.
- So an infrared telescope sees the stars inside the cloud.



Why put a telescope in space?

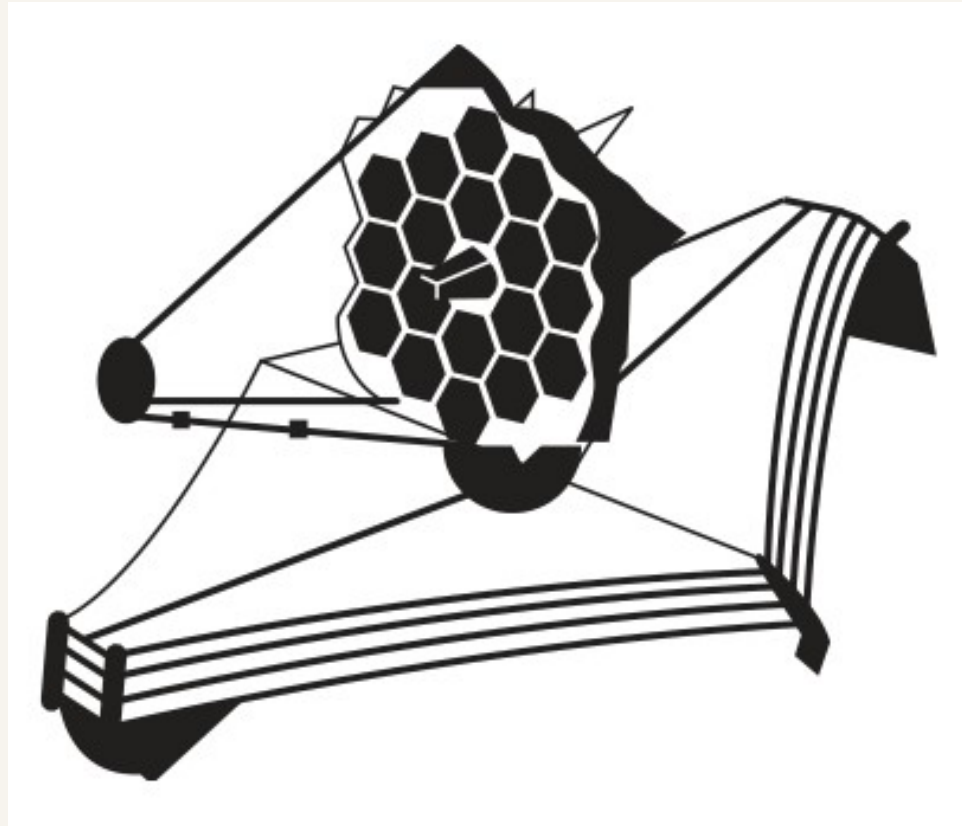
Booklet p.26-27

- The atmosphere absorbs some radiation, such as X-rays.
- Above it, every part of the spectrum can be detected.
- No weather, and no blurring from the air.



The James Webb Space Telescope

Booklet p.26, Task 2



Webb detects mainly infrared; Hubble mainly visible light.

The Pillars of Creation

Booklet p.26, Task 2 — same pillars, two telescopes



Hubble — visible light

NASA, ESA/Hubble and the Hubble Heritage Team (CC BY 4.0)



Webb — infrared

NASA, ESA, CSA, STScI; J. DePasquale, A. Koekemoer, A. Pagan (STScI) (CC BY 4.0)

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A dust cloud hides a group of new stars.
Which telescope is most likely to see them?

- A** A visible-light telescope
- B** An infrared telescope
- C** An X-ray telescope on the ground
- D** Any telescope with a bigger mirror
- E** None — the stars cannot be seen.

Answer B

Show me — telescopes

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Which part has the shortest wavelength?
- 2 Which part does James Webb mainly detect?
- 3 Why can X-ray telescopes not work on the ground?
- 4 Name a part of the spectrum that reaches the ground.
- 5 Order by wavelength: X-rays, radio, visible.

Gamma rays

Infrared

The atmosphere absorbs
X-rays

Visible light

Radio, visible, X-rays

Task 2 – answer the questions

Booklet p.26

Answer the five questions about
telescopes and the spectrum.

Booklet p.26, Task 2

Task 2 – Answers

ANSWERS

Booklet p.26

Question	Answer
Speed of microwaves	$3.0 \times 10^8 \text{ m s}^{-1}$
Why different parts?	each shows us different information
Hubble mainly uses	visible light
James Webb mainly uses	infrared
The two pictures	Webb sees stars inside the dust

Task 3 – a Webb discovery

Booklet p.26

Pick a recent James Webb discovery
and summarise it in a paragraph.

Booklet p.26, Task 3



[NASA Webb](#)

Demo — see the infrared

TRY THIS

Point a TV remote at your eyes and press a button.
Nothing. Now look at it through a phone camera.

The camera's sensor detects infrared; your eyes do not.
Which telescope does that make the phone?

What a 3-mark answer shows

Open-ended questions

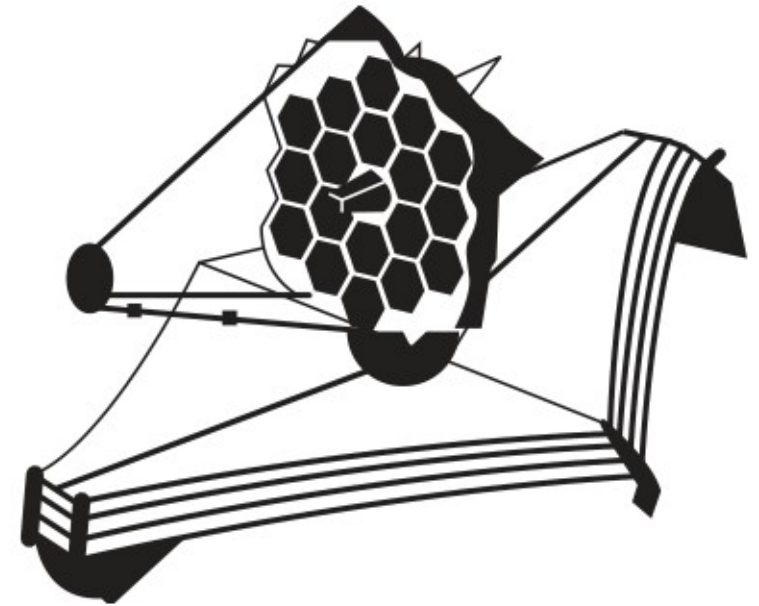
- 3 marks: good understanding – the principles, a relationship or equation, applied. It need not be complete.
- 2 marks: reasonable understanding – relevant statements.
- 1 mark: limited understanding – a little of the physics.
- 0 marks: nothing relevant, or just repeating the question.

Open-ended practice

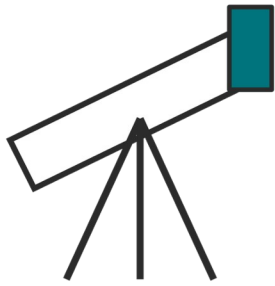
In your jotter (3 marks)

A pupil says that a telescope on a mountain top is just as good as one in space.

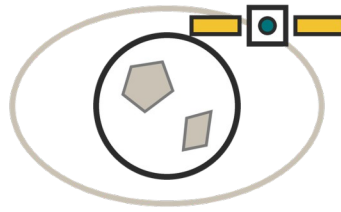
Using your knowledge of physics, comment on this.



Careers – where this physics goes



Astronomer



Satellite engineer



Data scientist

- Every space telescope needs people to build it, fly it and read it.
- UKube-1, Scotland's first satellite, was built in Glasgow (2014).

Checkpoint

- 1 List the electromagnetic spectrum, longest wavelength first.
- 2 Why do astronomers use more than visible light?
- 3 Give one advantage of putting a telescope in space.

Radio to gamma rays

Each part shows something different

No atmosphere in the way

Exam practice

Past paper and specimen paper

- National 5 Physics 2024, Section 2, question 6(a). 1 mark.
- 2023, Section 2, question 5(a). 1 mark. Space-based telescopes.
- Specimen paper, Section 1, question 15. 1 mark. The order of the spectrum.
- Then booklet p.27, Task 4.

Review 2

Booklet p.27

Check your booklet is up to date.
Test yourself on the definitions.
Sit Space Practice Test A on the website,
then Practice Test B. Compare scores.
Fill in your learning log, and revise.

Booklet p.27, Review

LOOKING AHEAD

The Space Test

Practice Tests A and B first — then the real thing.

Plenary

- 1 The big question: why look at one object in every kind of light?
- 2 Which part of the spectrum has the longest wavelength?
- 3 What do all parts of the spectrum have in common?
- 4 Which part does the James Webb telescope mainly use?
- 5 Why are some telescopes put into orbit?

Each kind shows something different

Radio waves

They travel at 3.0×10^8
 m s^{-1}

Infrared

No atmosphere in the way

Plenary

EXTRA PRACTICE

- 1 Which part lies between visible light and X-rays?
- 2 Which part lies between radio and infrared?
- 3 Which part does Hubble mainly use?
- 4 Name a part blocked by the atmosphere.
- 5 Why can infrared show what visible light cannot?

Ultraviolet

Microwaves

Visible light

X-rays

It passes through dust

Extension

- 1 Suggest why radio telescopes can work on cloudy days.
- 2 Why must an infrared telescope be kept very cold?
- 3 Suggest why a dust cloud that hides stars is still worth studying.

Extension

EXTRA PRACTICE

- 1 Why is a space telescope more expensive to fix than one on the ground?
- 2 Suggest why X-rays tell us about very hot objects in space.
- 3 Why might two pictures of one object look completely different?

Course specification

N5 Physics course specification v4.0, p.8 – Cosmology

- Awareness of the use of the whole electromagnetic spectrum in obtaining information about astronomical objects.